BOTANY HUB MULTAI

B.Sc. I BOTANY — MINOR PAPER II

DIVERSITY OF PLANTS

UNIT IV — Gymnosperms — Cycas and Pinus

DETAILED TEACHER-STYLE NOTES

Definitions • Complete explanations • Classification tables • Life cycles • Flowcharts • Diagrams • Examples • Economic importance • Exam points

Prepared strictly from the uploaded 2025–26 Minor Paper II syllabus.

UNIT IV — GYMNOSPERMS

This unit requires general account, classification, morphology, anatomy, reproduction and life cycle of Cycas and Pinus, plus economic importance. ■filecite■turn11file0■L71-L76■

Definition — Gymnosperms: Seed plants in which ovules and seeds are not enclosed within a typical closed ovary/fruit as in angiosperms; reproductive structures are commonly cones or specialized strobili.

General characteristics

- Dominant sporophyte is a woody vascular plant in many members.
- Ovules are exposed on megasporophylls or cone scales.
- Heterospory is present: microspores and megaspores are produced.
- Pollination is commonly by wind.
- Fertilization and seed development occur without formation of a typical angiosperm fruit.

Classification — broad textbook outline

Major group	Representative genera
Cycadophyta/Cycadopsida	Cycas
Ginkgophyta	Ginkgo
Coniferophyta/Coniferopsida	Pinus and other conifers
Gnetophyta/Gnetopsida	Gnetum, Ephedra

Cycas — morphology

Cycas has a stout, usually unbranched trunk crowned by large pinnate leaves. It is dioecious: male and female reproductive structures occur on separate plants. Coralloid roots may occur and are associated with cyanobacterial symbionts.

Cycas — anatomy and reproduction

A typical study answer should discuss protective tissues, cortex and vascular organization of the organ specified in practical/theory diagrams. Reproduction involves microsporophylls and megasporophylls, formation of pollen and ovules, pollination, fertilization and seed development.

Simplified reproductive sequence in Cycas

Male plant: microspores /pollen → **Pollination** → **Female ovule** → **Male gamete reaches egg** → **Fertilizati on** → **Embryo + seed** → **New sporophyte**

Pinus — morphology

Pinus is an evergreen conifer with a woody trunk and needle-like leaves, commonly borne in fascicles. It is typically monoecious, with separate male and female cones on the same plant.

Pinus — anatomy and reproduction

Needle leaves show xerophytic adaptations such as reduced surface area, protective epidermal features and internal tissues suited to water conservation. Male cones produce pollen; female cones bear ovules on ovuliferous scales. Wind pollination is common, followed by fertilization and seed formation.

Simplified life cycle of Pinus

Cycas vs Pinus

Feature	Cycas	Pinus
Habit	Palm-like crown with stout trunk	Coniferous tree
Sex distribution	Usually dioecious	Usually monoecious
Leaves	Large pinnate leaves	Needle-like leaves
Female structure	Megasporophylls not forming a typical compact cone	Woody female cone
Pollination	Generally wind	Generally wind

Economic importance

• Timber and paper industry, especially from conifers.

• Resins, turpentine and related products.

• Ornamental and landscape uses.

• Seeds and edible products in selected gymnosperms.

• Scientific importance for understanding seed-plant evolution.

Exam focus: Do not confuse gymnosperm 'naked seed' with literally uncovered seed at every stage. The key comparison is that ovules are not enclosed within an angiosperm-type ovary that develops into a fruit.